1. Methodology

The methodology used for the Inventory Procurement & Reorder System is the **Waterfall Methodology**.

Waterfall is a step-by-step methodology in which each phase must be completed before moving to the next phase.

The project is divided into four main phases:

1. **Planning:** Identifying the problem, project objectives, and project requirements.
2. **Analysis:** Studying the current inventory system and collecting information about stock management, suppliers, and reorder processes.
3. **Design:** Designing the system structure, processes, and reports.
4. **Implementation:** Preparing the final system documentation and checking that all project requirements are covered.

Reason for choosing Waterfall:

Waterfall is suitable for this project because its requirements are clear, and the project can be divided into organized phases with specific tasks and deadlines.

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2. Time Estimation

Time estimation determines how much time is required to complete each phase of the project.

The project follows these estimated percentages:

Phase	Percentage	Estimated Time
Planning	15%	11 days
Analysis	20%	16 days
Design	35%	25 days
Implementation	30%	22 days
Total	**100%**	**74 days**

Explanation

- **Planning:** 11 days to identify the problem, objectives, and feasibility.
- **Analysis:** 16 days to collect and analyze information about the current inventory system.
- **Design:** 25 days to prepare the project diagrams and define the proposed system processes.
- **Implementation:** 22 days to complete, organize, and review the final project documentation.

Note: These are estimated durations based on the project schedule. The actual time may differ depending on the work required.

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3. Case Study

Inventory Procurement & Reorder System

Current Situation

A small retail store manages its inventory manually using notebooks or spreadsheets. The store sometimes faces problems such as incorrect stock quantities, product shortages, delayed purchasing, and difficulty tracking suppliers and purchase orders.

Problem

The store does not have an organized way to monitor stock levels and identify when products need to be reordered. This may cause delays and affect the availability of products.

Proposed Solution

The Inventory Procurement & Reorder System will help the store organize its inventory information and monitor product quantities.

The system will:

1. Record product information.
2. Track the available quantity of each product.
3. Define a reorder level for every product.
4. Identify products that need to be reordered.
5. Store supplier information.

6. Organize purchase order information.
7. Generate inventory and reorder reports.

Example

A store has 8 keyboards in stock, and the reorder level is 10 keyboards.

Since the available quantity is below the reorder level, the system identifies the keyboard as a product that needs to be reordered.

The store can then prepare a purchase order to request additional keyboards from the supplier.

Expected Benefits

- Reducing product shortages.
- Improving inventory accuracy.
- Making purchasing decisions easier.
- Organizing supplier information.
- Saving time when checking stock levels.